

KARTHIK NAKKALA

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🌐 LinkedIn 🐙 GitHub 🌐 Portfolio

EDUCATION

Indian Institute of Technology Madras 2024 - 2027
B.S. in Data Science and Applications CGPA: 7.44/10

IIIT Idupulapaya (Andhra Pradesh State Board) 2022
Senior Secondary (XII), Science Percentage: 90.10%

WORK EXPERIENCE

RIFT, Bengaluru Apr 2025 - Jul 2025
Junior Web Developer

- Engineered PharmaGuard AI for genomic analysis; integrated LLMs for CPIC guidelines interpretation.
- Architected real-time risk stratification systems with validated JSON schemas for clinical data.

Amazon India Hackathon Nov 2025
National Finalist - Top 5 (AWS Cloud AI Solutions)

- Developed a generative AI app using AWS PartyRock and Prompt Engineering in a 24-hour sprint.

PROJECTS

PharmaGuard (AI & Genomics) — [Link](#) Feb 2026

- Built an AI Clinical Decision Support platform using React/LLMs to prevent Adverse Drug Reactions.
- Visualized genetic phenotypes and risk assessments using Framer Motion for high-engagement UI.

Personal Portfolio Website — [Link](#) Jan 2026

- Architected high-performance brand platform with React/Tailwind; achieved high accessibility scores.

Multi-View Project Tracker — [Link](#) Mar 2026

- Built a project management tool with three switchable views (Kanban, List, Timeline) sharing a single Redux state.
- Implemented custom drag-and-drop from scratch (mouse/touch, placeholder, snap-back) without any external libraries.
- Engineered virtual scrolling for 500+ tasks with fixed row heights and dynamic DOM rendering; achieved 100 Lighthouse performance score.
- Integrated URL-synced filters (status, priority, assignee, date range) and simulated real-time collaboration with avatar indicators.
- Tech stack: React, TypeScript, Redux Toolkit, Tailwind CSS, custom hooks.

Data Structures and Algorithms Feb 2025 - Present

- Solved 300+ problems (LeetCode, GFG, HackerRank) covering DP, Trees, and Graph Theory.

TECHNICAL SKILLS

- Languages:** C, C++, Java, Python, JavaScript (ES6+), TypeScript, SQL
- Frontend:** React, Redux Toolkit, Tailwind CSS, HTML5, CSS3, Tailwind css, jQuery, Framer Motion
- Backend:** Node.js, Express.js, MongoDB
- Tools:** Git, GitHub, VS Code, Vercel, Netlify, AWS PartyRock